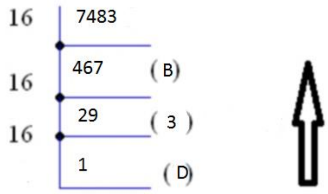
# Course Code: EET206

# Course Name: DIGITAL ELECTRONICS

## PART A

**1. Convert**

**a) (7483)10 into hexadecimal**




**(Hexadecimal)**

Ans = $(1D3B)_{16}$

$$(7483)_{10} = (1D3B)_{16}$$

**b) 1 1001 0100 into Gray Code**

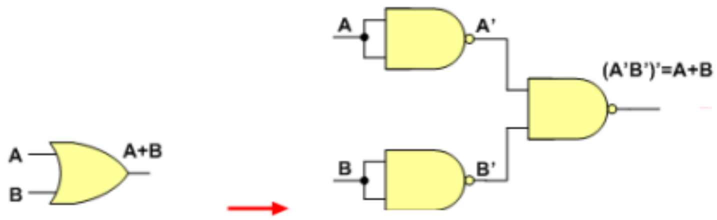
Gray Code = 101011110

**2. Realize NOT, AND and OR gate using NAND gates only**

NOT Gate


AND Gate

OR Gate




**3. State and explain De Morgan's theorem**

**The complement of a product of variables is equal to the sum of the complements of the variables.**

Stated another way,

**The complement of two or more ANDed variables is equivalent to the OR of the complements of the individual variables.**

The formula for expressing this theorem for two variables is

$$\overline{XY} = \overline{X} + \overline{Y}$$

**The complement of a sum of variables is equal to the product of the complements of the variables.**

Stated another way,

**The complement of two or more ORed variables is equivalent to the AND of the complements of the individual variables.**

The formula for expressing this theorem for two variables is

$$\overline{X + Y} = \overline{X}\,\overline{Y}$$

**4. Prove that A+A'B=A+B**

$$A+A'B=A(1+B)+A'B=A+AB+A'B=A+(A+A')B=A+B$$

**5. Draw a 4-bit gray to binary code converter circuit**

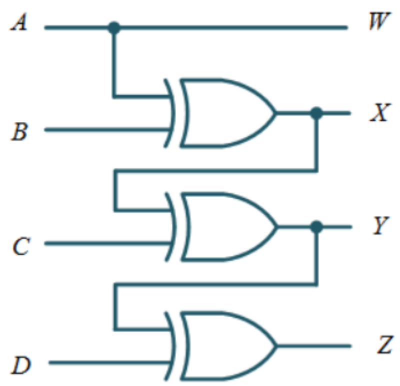
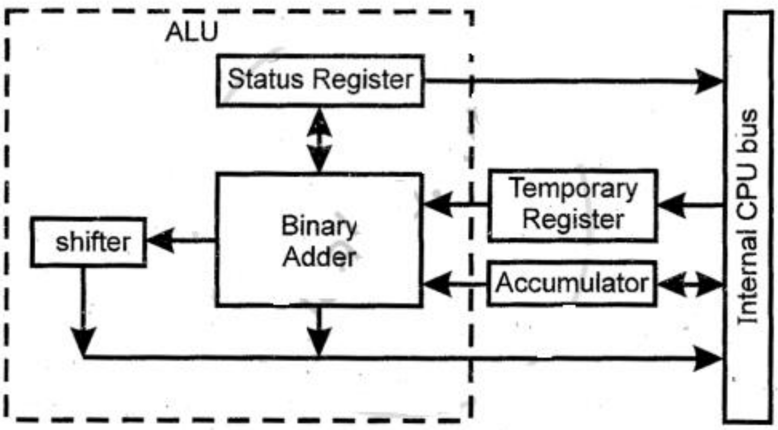



**Figure** : *Gray to Binary Code Converter logic diagram.*

**6. Draw the block diagram of ALU.**



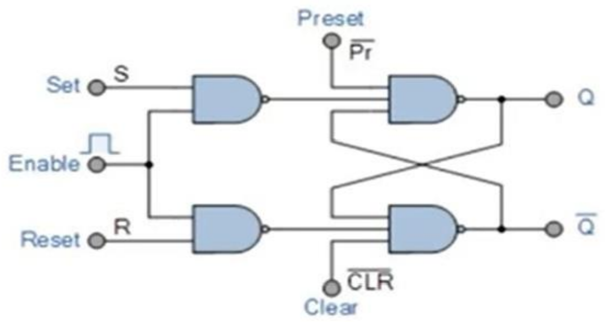

**7. Explain Preset and Clear inputs of a flip-flop.**

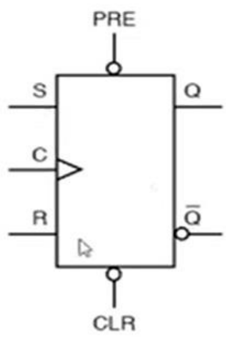
The normal data inputs of flipflop (D, T,S and R or J and K) are called synchronous inputs since they have effect on the outputs of the flip flop when the clocked pulse is present.

Flip flop can also have other inputs called Preset (or SET) and clear that can be used for setting the flip flop to 1 or resetting it to 0 by applying the appropriate signal to the Preset and Clear inputs. These inputs can change the state of the flip flop regardless of synchronous inputs or the clock. Both are active low signals generally. They are called asynchronous inputs or overriding inputs.

They are normally labelled preset (PRE) and clear (CLR), or direct set (SD) and direct reset (RD) by some manufacturers.

An active preset input makes the Q output HIGH (SET). An active clear input makes the Q output LOW (RESET).



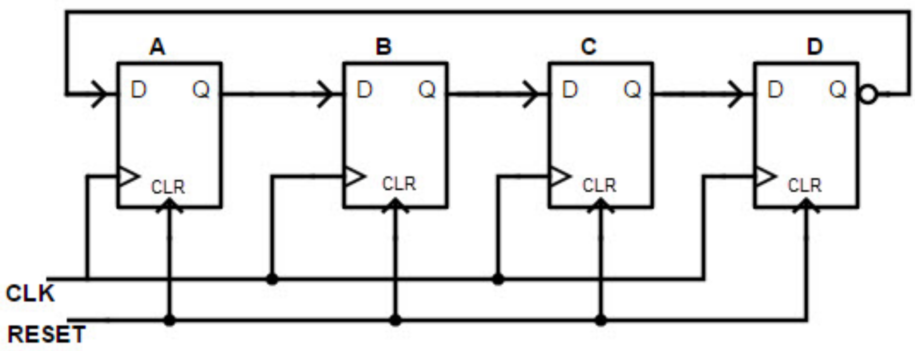




| Preset | Clear | Qn |
|---|---|---|
| 0 | 0 | Not Used |
| 0 | 1 | 1 |
| 1 | 0 | 0 |
| 1 | 1 | FF will perform normally |

Working of SR Flipflop with Preset and clear inputs

- If preset = 0, it will set the flip flop output Q as 1 without considering S,R and clock inputs.
- If clear = 0, it will set the flip flop output Q as 0 without considering S,R and clock inputs.
- Preset and clear will be very useful to set and reset the flipflop at the initial stage.
- When preset= clear =0, it is not possible to make Q as 1 and 0.So it is not used.
- When preset= 0 and clear =1, we can make Q as 1.
- When preset= 1 and clear =0, we can make Q as 0.
- When preset=clear =1, Flipflop will perform normally according to S,R and clock inputs.

**8. Draw the logic diagram of a 4 bit Johnson counter and explain its working**




The **johnson counter circuit diagram** is the cascaded arrangement of 'n' flip-flops. In such design, the output of the proceeding flip-flop is fed back as input to the next flip-flop. For example, the inverted output of the last flip-flop '$\overline{Q}$n' is fed back to the first flip-flop in the sequence bit pattern. The counter registers cycles in a closed-loop i.e circulates within the circuit.
Consider the 4-bit Johnson counter, it contains 4 D flip-flops, which is called 4-bit Johnson counter. It has preset and clear pins to initialize or start and reset the counted.

Reset pin acts as an on/off switch. So, the flip-flops can be enabled by clicking the Reset switch.

CLK pin is used to observe the changes in the output of the flip-flops.

Standard 2,3 and 4 stages johnson counters are used to divide the frequency of clock signals with the help of varying feedback connections. For example, a 3-stage johnson counter can be used as a 3-phase and 120 degrees phase shift square wave generator. 5-stage Johnson counter is used as a synchronous decade counter (CD4017) or divider circuit. 2-stage acts as a quadrature oscillator or generator that produces individual output signals of 90 degrees each concerning the input signal.

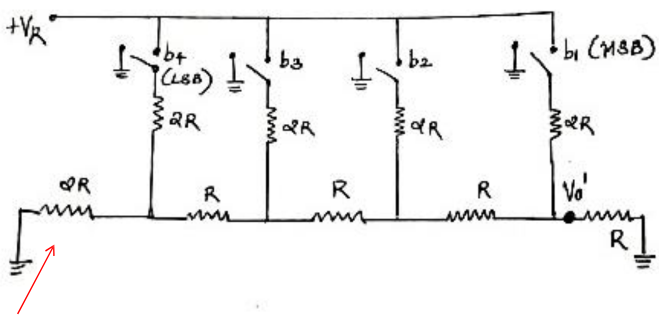
9. Draw and explain R-2R ladder type DAC

A wide range of resistors are required in binary weighted type DAC. This can be avoided by using R-2R ladder type DAC where only two types of resistors, R and 2R are required. Typical values of R ranges from 2.5 kilo-ohms to 10 kilo-ohms.So IC fabrication is easy.

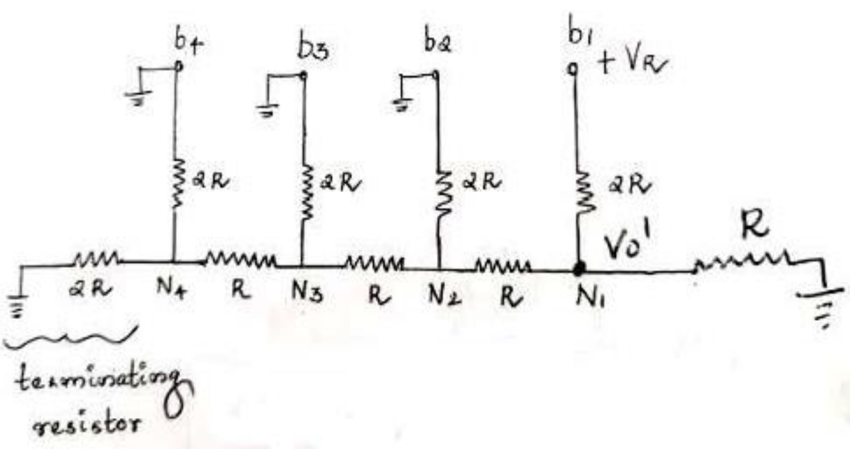
Here it uses a ladder network containing series and parallel combinations of values of R and 2R .It also uses op amp in the negative feedback configuration to perform analog to digital conversions.

Given below is an R-2R ladder arrangement for 4-bit digital data. V0' is the output voltage of the ladder arrangement. Kindly note that there is also a terminating resistor 2 R.

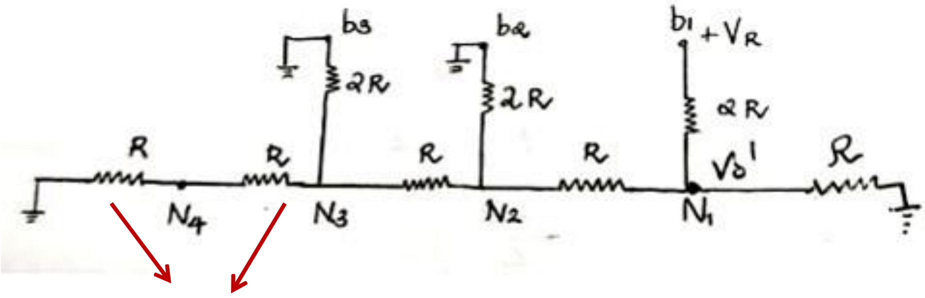
Each bit corresponds to switch. If bit is high, the corresponding switch is connected to the inverting input of op- amp. If bit is low, switch is connected to the ground.



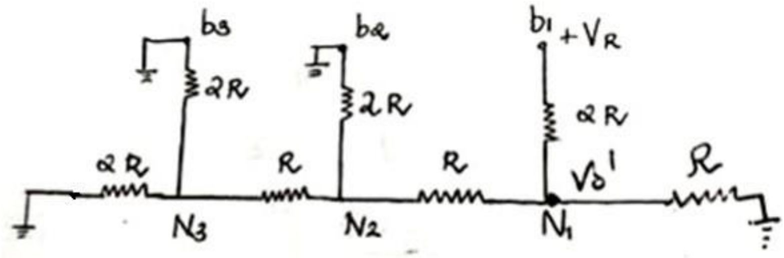

- Assume that the 4-bit binary input is 1000, i.e. b1b2b3b4 = 1000 (MSB is b1). The b1 switch connects to VR and b2, b3 and b4 connects to ground as shown below



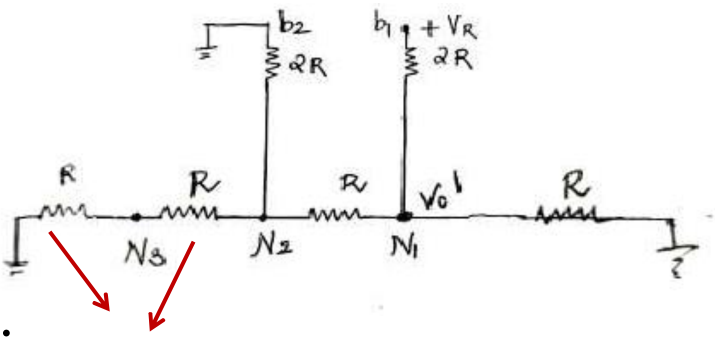
- The terminating resistor 2R and the resistor 2R connected to b4 are in parallel, so they can be combined at node N4 to form an equivalent resistance R as shown in the next figure



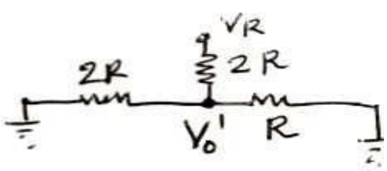

This R and R can be combined to form series equivalent resistance 2R.

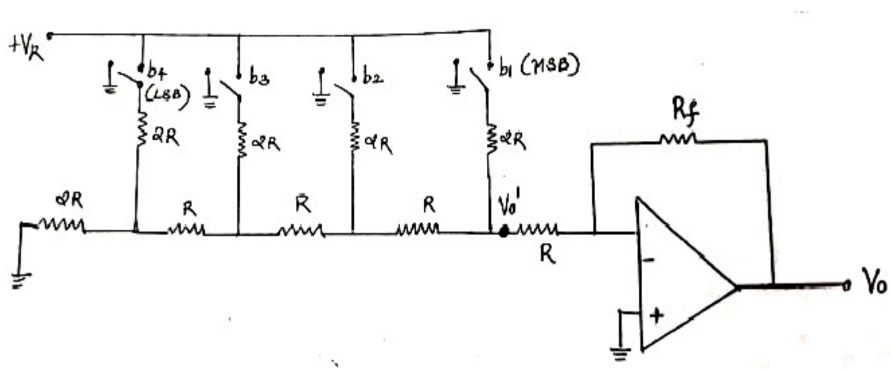



The resistor 2R and the resistor 2R connected to b3 are in parallel, so they can be combined at node N3 to form R




- This R and R can be combined to form series equivalent resistance 2R.






10. Differentiate between PLA and PAL

**PLA**

- It stands for Programmable Logic Array.
- Its speed is lesser in comparison to PAL.
- It is highly complex.
- It is expensive.
- It is not available easily.
- It is used less in comparison to PAL.

**PAL**

- It stands for Programmable Array Logic.
- It has a higher speed in comparison to PLA.
- It is less complex.
- It is inexpensive.
- It is more readily available in comparison to PLA.
- It is used more in comparison to PLA.

PART B

11. (a) Subtract 46 from 99 using 1's complement and 2's complement methods. Compare both methods.

**1's Complement Method**

99 = 0 1 1 0 0 0 1 1

\- 46 = 0 0 1 0 1 1 1 0

1's complement of 46 = 1 1 0 1 0 0 0 1

```
   0 1 1 0 0 0 1 1 +
   1 1 0 1 0 0 0 1
 -------------------------
 1 0 0 1 1 0 1 0 0
                 1
-----------------------------
   0 0 1 1 0 1 0 1
-----------------------------
```

**2's Complement Method**

99 = 0 1 1 0 0 0 1 1

\- 46 = 0 0 1 0 1 1 1 0

1's complement of 46 = 1 1 0 1 0 0 0 1

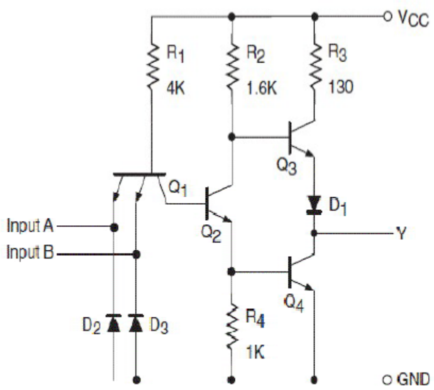
2's complement of 46 = 1 1 0 1 0 0 1 0

```
   0 1 1 0 0 0 1 1 +
   1 1 0 1 0 0 1 0
 -------------------------
 1 0 0 1 1 0 1 0 1
```

(b) Draw and Explain TTL NAND gate implementation

# TTL NAND Gate- Internal Diagram

- Here A and B are inputs and Y is the output.
- Diodes 2 and 3 are used to protect Q1 from unwanted signals.
- Input transistor Q1 is the multiple emitter transistor and Q3 act as emitter follower AND Q2 act as phase splitter.
- Transistor Q3 and Q4 make a totempole arrangement.
- Diode D1 ensures that Q3 and Q4 do not conduct simultaneously.




# Operation Of TTL NAND Gate

a) **A and B both Low: ( A=B =0)**

- Both Base –Emitter junctions of Q1 are forward biased. Hence D2 and D3 will conduct to force the voltage at point C to 0.7V.
- This voltage is insufficient to forward bias Base –Emitter junction of Q2. Hence Q2 remains OFF. Therefore its collector voltage rises to VCC. Also Q4 does not get required base drive signal so Q4 is also OFF.
- Q3 gets sufficient base drive signal so all current flows to the base of Q3.As Q3 is operating in emitter follower mode, output Y will be pulled up to high voltage Y= 1.

b) **Either A or B Low: (A= 1 ,B= 0) or (A= 0 ,B= 1)**

- If any one input is connected to ground with other left open or connected to VCC the corresponding diode (D2 or D3) will conduct. This will pull down voltage at C to 0.7V.
- This voltage is insufficient to turn on Q2 so it remains OFF. So collector voltage of Q2 will be equal to VCC. This voltage acts as base voltage for Q3.
- As Q3 acts as an emitter follower, output Y will be pulled to VCC. Thus Y= 1.

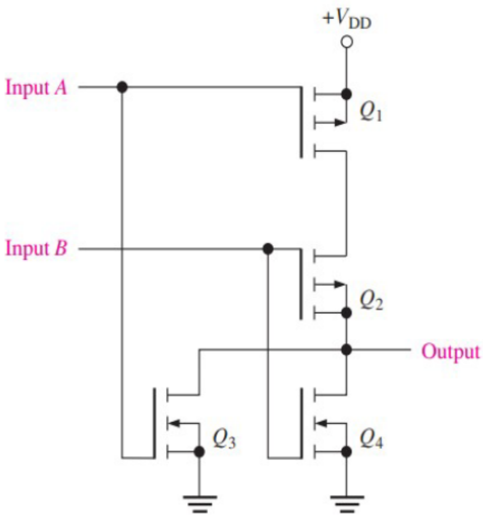
c) **A and B both High: (A=B =1)**

- If both A and B are connected then both diodes D2 and D3 will be reverse biased and do not conduct. So no current flows to Q1.
- But Collector Base junctions of Q1 is forward biased and thus current is supplied to the base of transistor Q2 via R1. That makes Q2 ON.
- As Q2 conducts, current flows from emitter of Q2 to the base of Q4.Thus Q4 will be ON.
- Collector current of Q2 flows through R2 and this produces a voltage drop across it which reduces the voltage at the collector of Q2 that makes Q3 OFF.
- Since Q4 is ON, the output voltage Y will be pulled down to low. Thus Y = 0.

**12.** (a) **With neat diagram, explain the operation of CMOS NOR gate.**

# CMOS NOR gate – Internal Diagram

- Q1 and Q2 are P MOS transistors connected in series.
- Q3 and Q4 are N MOS transistors connected in parallel.




| $A$ | $B$ | $Q_1$ | $Q_2$ | $Q_3$ | $Q_4$ | $X$ |
|---|---|---|---|---|---|---|
| L | L | S | S | C | C | H |
| L | H | S | C | C | S | L |
| H | L | C | S | S | C | L |
| H | H | C | C | S | S | L |

C = cutoff (off)
S = saturation (on)
H = HIGH
L = LOW

## Operation of CMOS NOR Gate

- When both inputs are LOW, Q1 and Q2 are on, and Q3 and Q4 are off.
- As a result, the output is pulled HIGH through the on resistance of Q1 and Q2 in series.
- When input A is LOW and input B is HIGH, Q1 and Q4 are on, and Q2 and Q3 are off. The output is pulled LOW through the low on resistance of Q4 to ground.
- When input A is HIGH and input B is LOW, Q1 and Q4 are off, and Q2 and Q3 are on. The output is pulled LOW through the on resistance of Q3 to ground.
- When both inputs are HIGH, Q1 and Q2 are off, and Q3 and Q4 are on. The output is pulled LOW through the on resistance of Q3 and Q4 in parallel to ground.

**13. (a) Design a full subtractor circuit using basic gates.**

| Minuend (A) | Subtrahend (B) | Borrow In ($B_{in}$) | Difference (D) | Borrow Out ($B_o$) |
|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 0 |
| 0 | 0 | 1 | 1 | 1 |
| 0 | 1 | 0 | 1 | 1 |
| 0 | 1 | 1 | 0 | 1 |
| 1 | 0 | 0 | 1 | 0 |
| 1 | 0 | 1 | 0 | 0 |
| 1 | 1 | 0 | 0 | 0 |
| 1 | 1 | 1 | 1 | 1 |

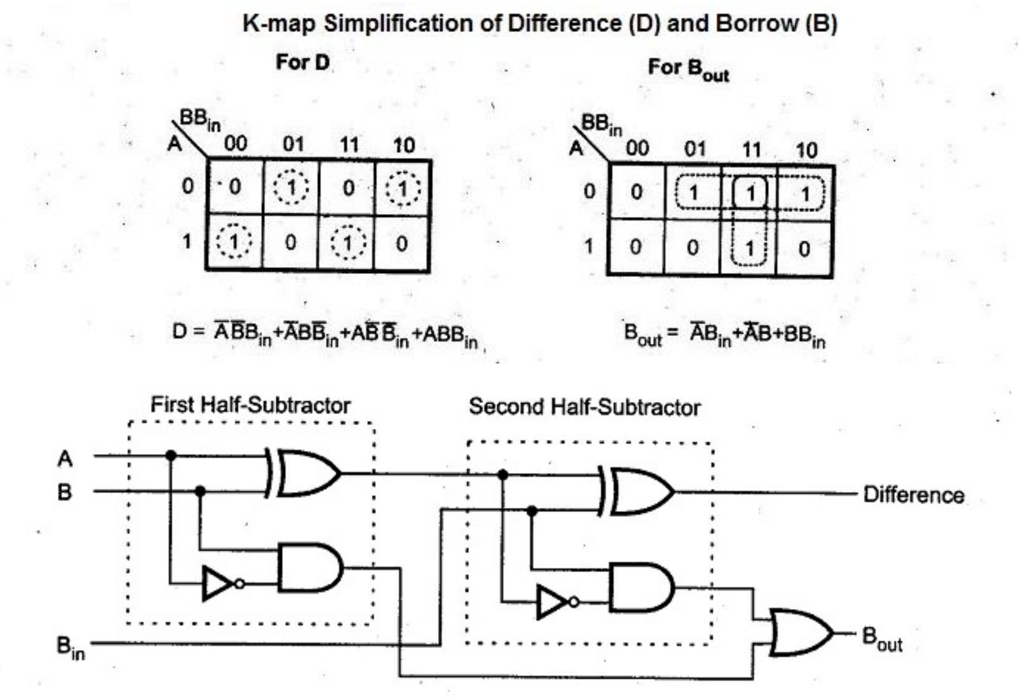


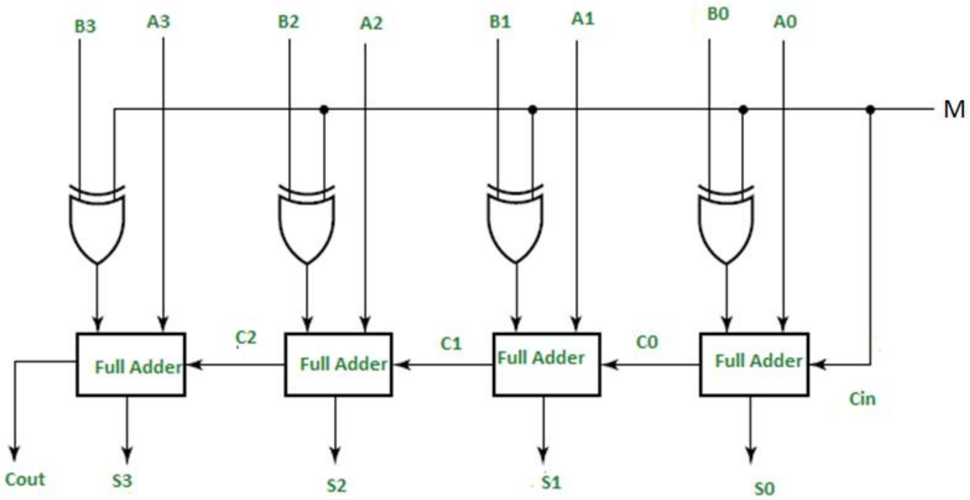

**Fig. 3.24 Implementation of a full-subtractor with two half-subtractors and an OR gate**

(b) Reduce the expression using K map, F(A,B,C,D) =∑m(6,7,8,10,11,15) + d(0,2,3,4,5,9,14)

| AB \ CD | 00 | 01 | 11 | 10 |
|---|---|---|---|---|
| 00 | X | | X | X |
| 01 | X | X | 1 | 1 |
| 11 | | | 1` | X |
| 10 | 1 | X | 1 | 1 |

$= A\bar{B} + C$

14. (a) Draw and explain four bit parallel adder/subtractor circuit.

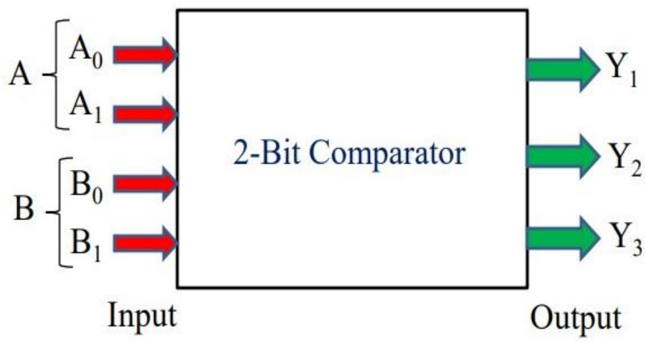



- Figure shows a 4 bit adder – subtractor circuit. Here the addition and subtraction operations are combined into one circuit with one common binary adder.
- This is done by including an X – OR gate with each full – adder.
- The mode input M controls the operation. When M= 0, the circuits is an adder, and when M = 1, the circuit becomes a subtractor. Each X –OR gate receives input M and one of the inputs of B.
- When M = 0, we have $B \oplus 0 = B$. The full adder receives the value of B, the input carry is 0 and the circuit performs A + B.

- When M = 1, we have B⊕ 1 = $\bar{B}$ $and\ C1 = 1.$ The B inputs are complimented and a 1 is added though the input carry. The circuit performs the operation A plus the 2's compliment of B.

**15. (a) Realize a 2- bit comparator circuit**

- A comparator which is used to compare two binary numbers each of two bits is called a 2-bit magnitude comparator.
- Fig. 2 shows the block diagram of 2-Bit magnitude comparator.
- It has four inputs and three outputs.
- Inputs are $A_0$ ,$A_1$,$B_0$ and $B_1$ and Outputs are $Y_1$, $Y_2$ and $Y_3$




| INPUT | | | | OUTPUT | | |
|---|---|---|---|---|---|---|
| $A_1$ | $A_0$ | $B_1$ | $B_0$ | $Y_1$=A<B | $Y_2$=(A=B) | $Y_3$=A>B |
| 0 | 0 | 0 | 0 | 0 | 1 | 0 |
| 0 | 0 | 0 | 1 | 1 | 0 | 0 |
| 0 | 0 | 1 | 0 | 1 | 0 | 0 |
| 0 | 0 | 1 | 1 | 1 | 0 | 0 |
| 0 | 1 | 0 | 0 | 0 | 0 | 1 |
| 0 | 1 | 0 | 1 | 0 | 1 | 0 |
| 0 | 1 | 1 | 0 | 1 | 0 | 0 |
| 0 | 1 | 1 | 1 | 1 | 0 | 0 |
| 1 | 0 | 0 | 0 | 0 | 0 | 1 |
| 1 | 0 | 0 | 1 | 0 | 0 | 1 |
| 1 | 0 | 1 | 0 | 0 | 1 | 0 |
| 1 | 0 | 1 | 1 | 1 | 0 | 0 |
| 1 | 1 | 0 | 0 | 0 | 0 | 1 |
| 1 | 1 | 0 | 1 | 0 | 0 | 1 |
| 1 | 1 | 1 | 0 | 0 | 0 | 1 |
| 1 | 1 | 1 | 1 | 0 | 1 | 0 |

## K-Map for A<B:

| $A_1A_0 \backslash B_1B_0$ | 00 | 01 | 11 | 10 |
|---|---|---|---|---|
| 00 | 0 | 1 | 1 | 1 |
| 01 | 0 | 0 | 1 | 1 |
| 11 | 0 | 0 | 0 | 0 |
| 10 | 0 | 0 | 1 | 0 |

## K-Map for A=B:

| $A_1A_0 \backslash B_1B_0$ | 00 | 01 | 11 | 10 |
|---|---|---|---|---|
| 00 | 1 | 0 | 0 | 0 |
| 01 | 0 | 1 | 0 | 0 |
| 11 | 0 | 0 | 1 | 0 |
| 10 | 0 | 0 | 0 | 1 |

For A<B

$$Y_1 = \overline{A_1}\,\overline{A_0}\,B_0 + \overline{A_1}\,B_1 + \overline{A_0}\,B_1\,B_0$$

For A=B

$$Y_2 = \overline{A_1}\,\overline{A_0}\,\overline{B_1}\,\overline{B_0} + \overline{A_1}\,A_0\,\overline{B_1}\,B_0 + A_1A_0B_1B_0 + A_1\,\overline{A_0}\,B_1\,\overline{B_0}$$

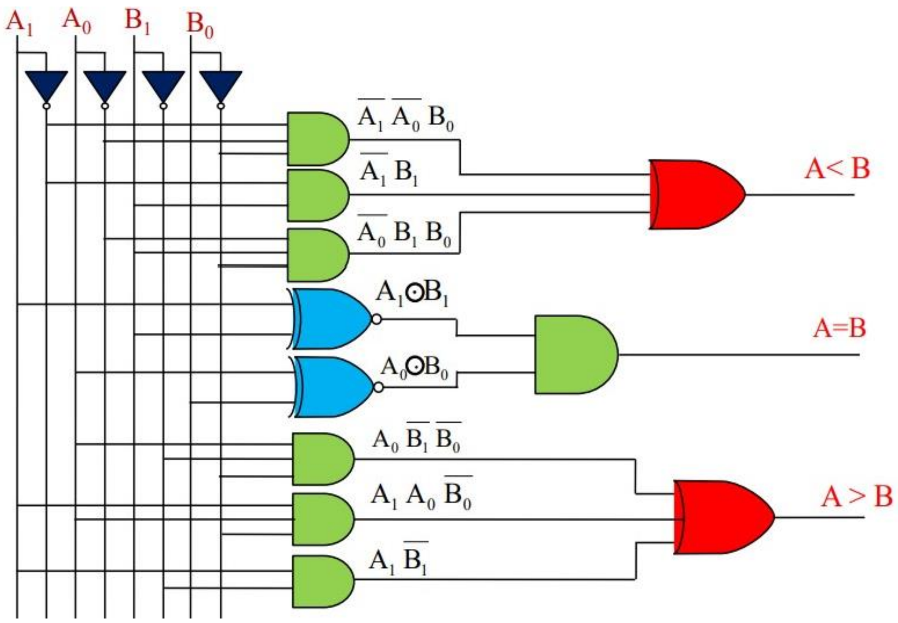
## K-Map For A>B

| $A_1A_0 \backslash B_1B_0$ | 00 | 01 | 11 | 10 |
|---|---|---|---|---|
| 00 | 0 | 0 | 0 | 0 |
| 01 | 1 | 0 | 0 | 0 |
| 11 | 1 | 1 | 0 | 1 |
| 10 | 1 | 1 | 0 | 0 |

$$Y_3 = A_0\,\overline{B_1}\,\overline{B_0} + A_1\,\overline{B_1} + A_1A_0\,\overline{B_0}$$

For A=B From K-Map

$$Y_2 = \overline{A_1}\,\overline{A_0}\,\overline{B_1}\,\overline{B_0} + \overline{A_1}\,A_0\,\overline{B_1}\,B_0 + A_1A_0B_1B_0 + A_1\,\overline{A_0}\,B_1\,\overline{B_0}$$
$$Y_2 = \overline{A_0}\,\overline{B_0}(\overline{A_1}\,\overline{B_1} + A_1B_1) + A_0B_0(\overline{A_1}\,\overline{B_1} + A_1B_1)$$
$$Y_2 = (\overline{A_1}\,\overline{B_1} + A_1B_1)\;(\overline{A_0}\,\overline{B_0} + A_0B_0)$$
$$Y_2 = (A_1 \odot B_1)(A_0 \odot B_0)$$




**16. (a) Differentiate decoder and encoder. Design a BCD to decimal decoder circuit**

| Decimal No | BCD CODE (Inputs) | | | | Decimal Outputs | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | W | X | Y | Z | D0 | D1 | D2 | D3 | D4 | D5 | D6 | D7 | D8 | D9 |
| 0 | 0 | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 1 | 0 | 0 | 0 | 1 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 2 | 0 | 0 | 1 | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 3 | 0 | 0 | 1 | 1 | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 0 |
| 4 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 0 |
| 5 | 0 | 1 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 0 |
| 6 | 0 | 1 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 0 | 0 |
| 7 | 0 | 1 | 1 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 0 |
| 8 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 0 |
| 9 | 1 | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 |
| INVALID | 1 | 0 | 1 | 0 | X | X | X | X | X | X | X | X | X | X |
| | 1 | 0 | 1 | 1 | X | X | X | X | X | X | X | X | X | X |
| | 1 | 1 | 0 | 0 | X | X | X | X | X | X | X | X | X | X |
| | 1 | 1 | 0 | 1 | X | X | X | X | X | X | X | X | X | X |
| | 1 | 1 | 1 | 0 | X | X | X | X | X | X | X | X | X | X |
| | 1 | 1 | 1 | 1 | X | X | X | X | X | X | X | X | X | X |

| WX \ YZ | 00 | 01 | 11 | 10 |
|---|---|---|---|---|
| 00 | $D_0$ | $D_1$ | $D_3$ | $D_2$ |
| 01 | $D_4$ | $D_5$ | $D_7$ | $D_6$ |
| 11 | $X$ | $X$ | $X$ | $X$ |
| 10 | $D_8$ | $D_9$ | $X$ | $X$ |

**Map for simplifying a BCD-to-decimal decoder**

$D_0 = W'X'Y'Z'$

$D_1 = W'X'Y'Z$

$D_2 = X'YZ'$

$D_3 = X'YZ$

$D_4 = XY'Z'$

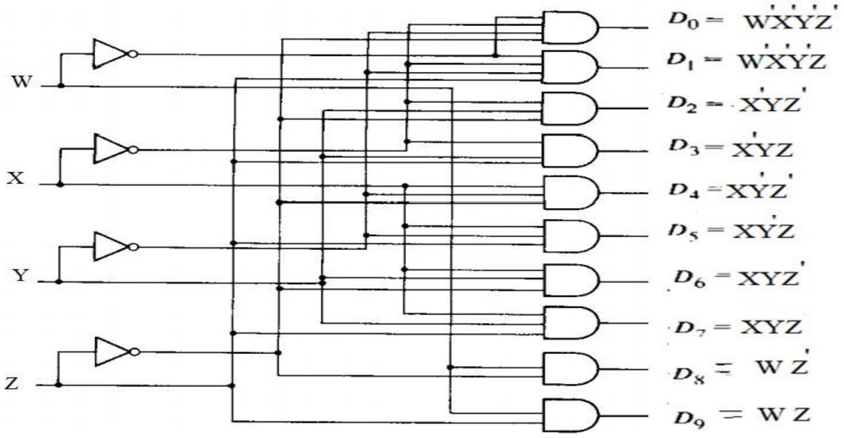
$D_5 = XY'Z$

$D_6 = XYZ'$

$D_7 = XYZ$

$D_8 = WZ'$

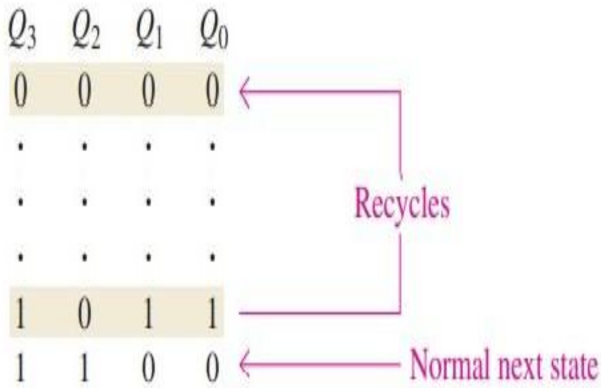
$D_9 = WZ$

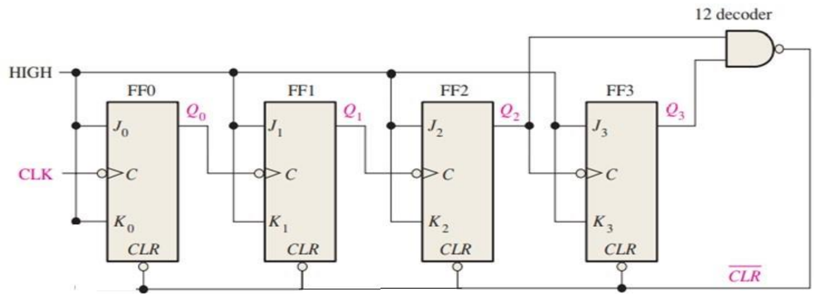


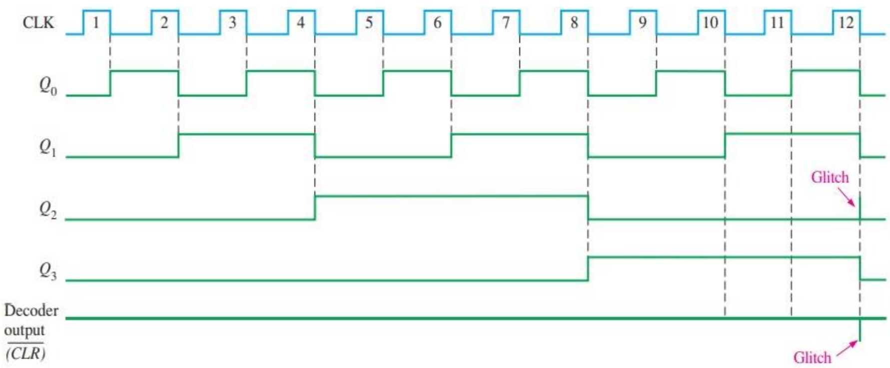
17. (a) Design a mod-12 asynchronous counter using J-K flip flops. Draw the timing diagram.

MOD 12 Counter requires 4 flipflops.

When the counter gets to its last state, 1011, it must recycle back to 0000 rather than going to its normal next state of 1100, as illustrated in the following sequence chart:



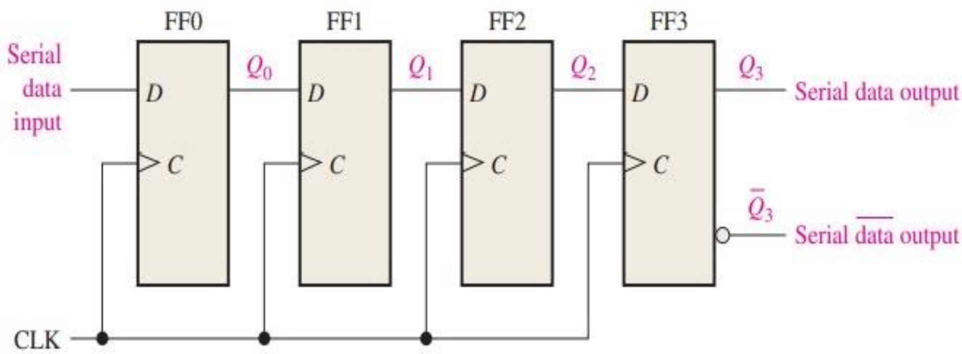



(b) Explain different types of shift registers.

- Shift registers consist of arrangements of flip-flops and are important in applications involving the storage and transfer of data in a digital system.
- A register has no specified sequence of states, except in certain very specialized applications. A register, in general, is used solely for storing and shifting data (1s and 0s) entered into it from an external source and typically possesses no characteristic internal sequence of states.
- A register is a digital circuit with two basic functions: data storage and data movement. The storage capability of a register makes it an important type of memory device.
- A register can consist of one or more flip-flops used to store and shift data.D flipflops are used for this purpose.
- The storage capacity of a register is the total number of bits (1s and 0s) of digital data it can retain. Each stage (flip-flop) in a shift register represents one bit of storage capacity; therefore, the number of stages in a register determines its storage capacity.

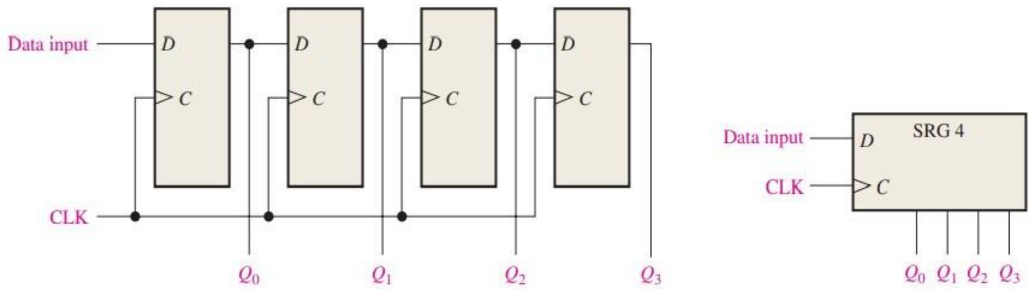
1. Serial In, Serial Out Shift Register (SISO)




- A 4-bit device implemented with D flip-flops. With four stages, this register can store up to four bits of data.
- Figure shows the entry of the four bits into the register , beginning with the least significant bit in a serial manner.
- The entry of the four bits 1010 into the register in Figure , beginning with the least significant bit. The register is initially clear. The 0 is put onto the data input line, making D = 0 for FF0. When the first clock pulse is applied, FF0 is reset, thus storing the 0.

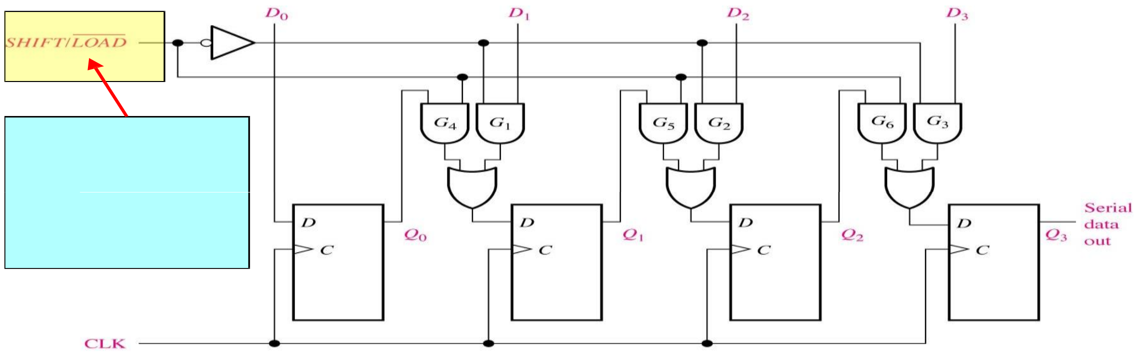
- Next the second bit, which is a 1, is applied to the data input, making D = 1 for FF0 and D = 0 for FF1 because the D input of FF1 is connected to the Q0 output. When the second clock pulse occurs, the 1 on the data input is shifted into FF0, causing FF0 to set; and the 0 that was in FF0 is shifted into FF1.
- The third bit, a 0, is now put onto the data-input line, and a clock pulse is applied. The 0 is entered into FF0, the 1 stored in FF0 is shifted into FF1, and the 0 stored in FF1 is shifted into FF2.

2. <u>Serial In, Parallel Out Shift register (SIPO)</u>




- Data bits are entered serially (least-significant bit first) into a serial in/parallel out shift register in the same manner as in serial in/serial out registers.
- The difference is the way in which the data bits are taken out of the register; in the parallel output register, the output of each stage is available.
- Once the data are stored, each bit appears on its respective output line, and all bits are available simultaneously, rather than on a bit- by-bit basis as with the serial output. 4 bit register is given below;

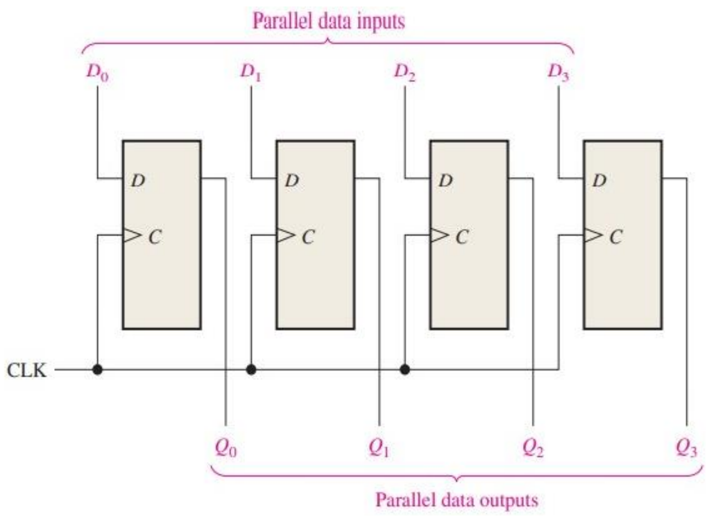
3. <u>Parallel in/serial out shift register (PISO)</u>




- 2 modes of operation – Load mode and Shift mode

- When SHIFT/LOAD is LOW, gates G1,G2 and G3 are enabled, allowing each data bit to be applied to the D input of its respective flip-flop. When a clock pulse is applied, the flip-flops with D = 1 will set and those with D = 0 will reset, thereby storing all four bits simultaneously.
- When SHIFT/LOAD is HIGH, gates G1,G2 and G3 are disabled and gates G4,G5 and G6 are enabled, allowing the data bits to shift right from one stage to the next. The OR gates allow either the normal shifting operation or the parallel data-entry operation, depending on which AND gates are enabled by the level on the SHIFT/LOAD input. Notice that FF0 ,does not require an AND/OR arrangement because there is no serial data in.

4-bit parallel in/serial out shift register (PISO)




- Data s are taken and entered in parallel form.
- The PIPO shift register is the simplest and fastest of the four configurations.
- Also known as buffer register or storage register.

18.(a) Design mod-14 synchronous counter using T-flip flop by explaining the steps in detail.

(b) Convert J-K flip-flop to T flip-flop.

1. Truth Table for T Flip-Flop

| Input | Outputs | |
|---|---|---|
| T | $Q_n$ | $Q_{n+1}$ |
| 0 | 0 | 0 |
| 0 | 1 | 1 |
| 1 | 0 | 1 |
| 1 | 1 | 0 |

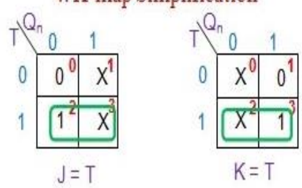
2. Excitation Table for JK Flip-Flop

| Outputs | | Inputs | |
|---|---|---|---|
| $Q_n$ | $Q_{n+1}$ | J | K |
| 0 | 0 | 0 | X |
| 0 | 1 | 1 | X |
| 1 | 0 | X | 1 |
| 1 | 1 | X | 0 |

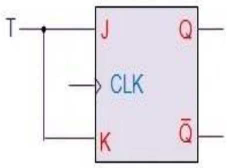
3. Conversion Table

| T | $Q_n$ | $Q_{n+1}$ | J | K |
|---|---|---|---|---|
| 0 | 0 | 0 | 0 | X |
| 0 | 1 | 1 | X | 0 |
| 1 | 0 | 1 | 1 | X |
| 1 | 1 | 0 | X | 1 |

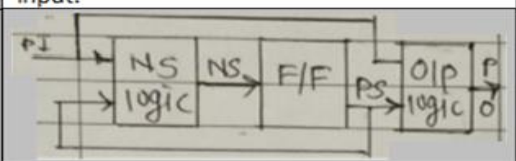
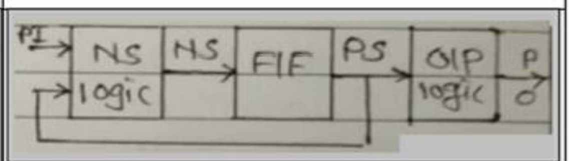
4. K-map Simplification




5. Circuit Design



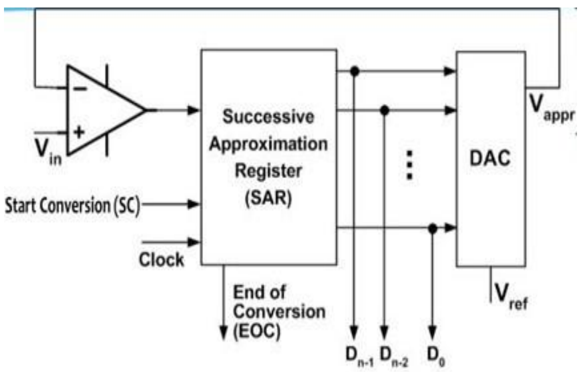

18. (a) Compare Mealy and Moore state machine models with example

| Sr. No | Mealy machine | Moore machine |
|---|---|---|
| 1 | The final output depends on the present state of memory element and the external input. | The final output only depends on the present state of memory element. |
| 2 |  |  |
| 3 | Output can change in between the clock edges if the external i/p changes. | The output changes only after the active clock edge. |
| 4 | It required less number states and there by less hardware to solve any problem. | It requires more number of states and hardware is more complex. |
| 5 | The disadvantage associate with the circuit is that if any input transients glitches are present, they are directly conveyed to the output. | The output remains stable over entire clock period and changes only when there occurs a state change at clock trigger band on i/p available at that time. |

**(b) Explain the working of (i) Successive approximation ADC and (ii) Flash type ADc**

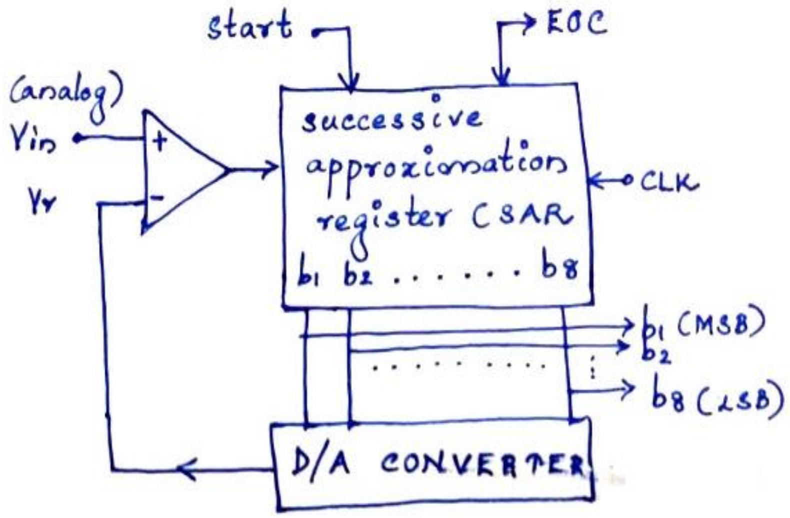
(i) Successive approximation ADC

- Successive approximation ADC consists of a comparator, Successive approximation Register (SAR) and DAC.
- A DAC is used to generate approximations of the input voltage.
- A comparator is used to compare Vin and V appr.
- In each cycle SAR finds one output using comparator.
- To start conversion, set SC= 1 . When conversion ends EOC = 1




8 Bit SAR ADC

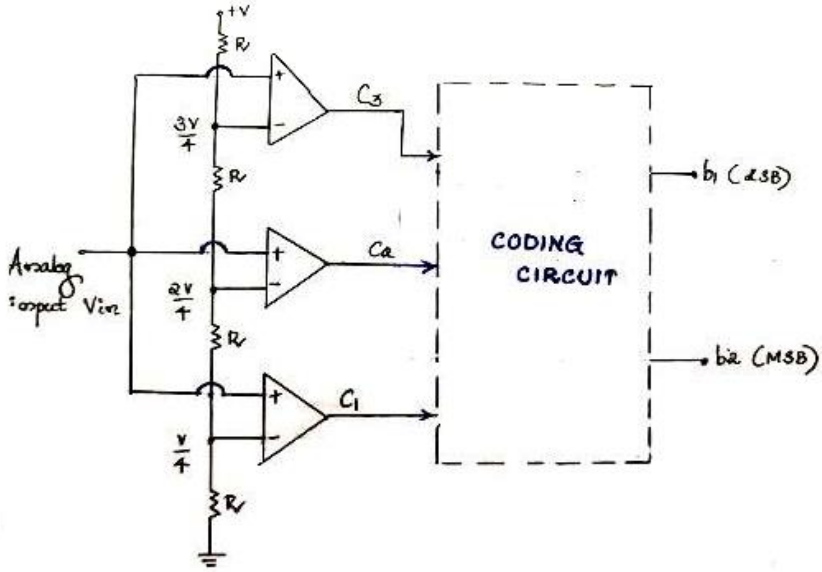
- The figure below shows a successive approximation type ADC that converts analog signals to 8-bit digital data.
- It has an 8-bit Successive Approximation Register (SAR) which checks whether each of the 8 bits is a 1 or a 0 by trial and error.




- When the start command is given, the first clock is applied and SAR sets b1 to 1 while all other bits are made 0. So the trail code in SAR become 10000000.
- This trial code is convert to analog voltage Vr and is compared with input Vin which the analog signal whose digital equivalent is to be found. If Vin is greater than Vr, it means that trail code 10000000 is less than the correct digital value of Vin. Comparator output becomes 1.
- In that case, b1 is retained as 1 and next bit b2 is set to 1, the trial code becomes 11000000. The trial code is converted to analog voltage Vr by DAC and is compared with Vin. . If Vin is greater than Vr, it means that trail code 11000000 is less than the correct digital value of Vin. Comparator output becomes 1.

(ii) Flash type ADc

- A simultaneous or flash type ADC is based on comparing an analog input voltage with a set of reference voltages .
- It is also known as parallel ADC.
- It is the fastest ADC and less complex compared to others.

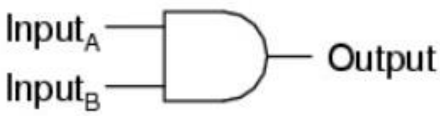



- The output of a comparator is in positive saturation state when the voltage at the non-inverting or +ve terminal is greater than at inverting or –ve terminal. Else comparator output is in negative saturation.
- When analog input Vin is less than V/4, comparator C1 output is 0. If Vin is less than V/4, it will be surely less than 2V/4 and 3V/4. SO comparators C2 and C3 output also is 0.
- When analog input Vin is between V/4 and V/2 comparator C1 output is 1 and C2 and C3 output is 0. When Vin is between V/2 and 3V/4, C1=C2=1 and C3=0. When Vin is between 3V/4 and V, C1=C2=C3=1

| Analog Input Voltage (Vin) | Comparator outputs | | | Digital Outputs | |
|---|---|---|---|---|---|
| | $C_1$ | $C_2$ | $C_3$ | $b_2$ | $b_1$ |
| $0 \le V_{in} \le V/4$ | 0 | 0 | 0 | 0 | 0 |
| $V/4 \le V_{in} \le V/2$ | 1 | 0 | 0 | 0 | 1 |
| $V/2 \le V_{in} \le 3V/4$ | 1 | 1 | 0 | 1 | 0 |
| $3V/4 \le V_{in} \le V$ | 1 | 1 | 1 | 1 | 1 |

- Since this is a 2-bit ADC, output digital data should have 2 bits, let this be b2 and b1. There are 4 combinations that b2 b1 can take.
- The coding circuit converts the comparator output to 2-bit digital output. That is the coding circuit is a combinational circuit made of gates that take C1C2C3 as input and outputs b2b1.

20 (b) Implement AND gate and a half adder circuit using Verilog.



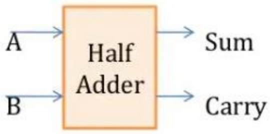

| A | B | Output |
|---|---|---|
| 0 | 0 | 0 |
| 0 | 1 | 0 |
| 1 | 0 | 0 |
| 1 | 1 | 1 |

Circuit Diagram With TT

```
module andgate (a, b, y);
input a, b;
output y;

assign y = a & b;

endmodule
```




Boolean Function

Sum = A^B

Carry = A.B

```
module half_adder (A , B , Sum , Carry);
  input A;             //first input to Half adder
  input B;             // Second input to Half Adder

  output Sum;          //output from the module
  output Carry;        //output from the module

  wire  Sum , Carry;   //wire declaration of the signal

  assign Sum = A ^ B;   // data flow representation of
    Sum
  assign Carry = A & B;    // data flow representation for
    Carry

endmodule
```